

NAMRATA LADDHA

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Astronautical Engineering | GPA: 3.73/4.0

August 2025-May 2027

Coursework: Spacecraft System Design, Spacecraft Power Systems, Ground Communications for Satellite Operations, Space Exploration Architectures Concept Synthesis Studio, Reliability of Space Systems,

Somaiya Vidyavihar University

Mumbai, India

Bachelors of Technology in Electronics Engineering, GPA: 3.76/4.0

July 2020-May 2024

- Honors in Artificial Intelligence, Computer Vision & Robotics

July 2020-Jan 2024

RELEVANT SKILLS

- Software: KiCAD, Altium, EasyEDA, LTSpice, Python, MATLAB, Simulink, Git, C++, Embedded C, Python, GNU Radio, Linux
- Technical Interest: Schematic Capture, PCB Designing, Power Electronics, Basic Electronic Circuits, Analog Integrated Circuits, and Analog Digital Communication
- Soft Skills: Efficient cross-functional communication, technical report writing, Cross-cultural teamwork, Detail oriented

EXPERIENCE

Laboratory for Autonomous Systems in Exploration and Robotics (LASER)

Los Angeles, CA

Graduate Student Researcher

May 2026-Present

- Leading electrical and software design of two NASA CADRE-class rovers as the experimental platform for multi-agent control algorithm research, architecting the power, RS485 motor bus, sensor fabric, and Jetson Orin Nano compute stack to support synchronized navigation, SLAM, and inter-rover coordination studies.

Space Engineering Research Center (Information Sciences Institute)

Los Angeles, CA

Graduate Student Researcher

September 2025-Present

- Developing and implementing hardware and software components for the Maveric's (CubeSat) electronics subsystem to meet mission requirements. Working on GNU Radio Flowgraphs to setup a reliable communication between CubeSat and Ground Station.

Liquid Propulsion Laboratory (USC)

Los Angeles, CA

Avionics & DAQ Engineer

September 2025-Present

- Design and development of Data Acquisition (DAQ) and instrumentation systems to capture sensor data and actuator control for real-time performance metrics on the liquid rocket engine test stand, proactively defining the sensor architecture to improve test safety

Manastu Space Technologies Private Ltd

Mumbai, India

Avionics Engineer

May 2024-April 2025

- Designed and delivered a flight-ready control electronics using STM32 and RS485 communication of propulsion system, deployed aboard ISRO's POEM Mission (PSLV C-60, Dec 2024)
- Performed FMECA, rigorous QA/QC, EMI/EMC, and environmental qualifications as per ISRO standards
- Lead Avionics Engineer for the IDEX mission with the Defense Space Agency, tasks: PCB Designing, Project Management, System Integration, Testing and Validation

Avionics research trainee

December 2023-May 2024

- Researched and developed low cost and low EMI, EPS (Electronic Power System) Solutions for propulsion module, reduced projected hardware costs by 15%
- Evaluated and tested various COTS components, validating functionality and space compliance for integration into propulsion system architecture

ACADEMIC PROJECTS

KJ Somaiya College of Engineering

Mumbai, India

Project Member

July 2024-December 2024

- Design and Development of a Model Rocket with Recovery System and Thrust Vectoring for Enhanced Performance and Stability. [Link](#)
- Payload Optimization Autonomous Drone: Designed and fabricated a custom flight control PCB, integrating the high-current motor drive circuits and sensor array for the real-time, embedded execution of constraint-based flight controls. [Link](#)